RSC Advances

RSC Publishing

PAPER

View Article Online
View Journal | View Issue

Published on 02 October 2013. Downloaded by University of Michigan Library on 19/10/2014 06:04:33.

Cite this: *RSC Adv.*, 2013, **3**, 23423

Spray-dried magnetic chitosan/Fe_3O_4/halloysite nanotubes/ofloxacin microspheres for sustained release of ofloxacin

Qin Wang, Junping Zhang* and Aiqin Wang

Received 25th July 2013
Accepted 1st October 2013

DOI: 10.1039/c3ra43874k

www.rsc.org/advances

A series of magnetic chitosan/Fe_3O_4/halloysite nanotubes/ofloxacin (CTS/Fe_3O_4/HNTs/OFL) microspheres was prepared by spray-drying for the sustained release of OFL. CTS was *in situ* crosslinked by glutaraldehyde in the spray-drying process. The magnetic microspheres were characterized by FT-IR, scanning electron microscopy and thermogravimetric analysis. The magnetic properties, encapsulation efficiency, drug loading and *in vitro* sustained release properties of the magnetic microspheres for OFL were also evaluated. The CTS/Fe_3O_4/HNTs/OFL microspheres show superparamagnetic properties and fast magnetic response. The Fe_3O_4 nanoparticles have a remarkable effect on the surface morphology of the microspheres compared with HNTs. The encapsulation efficiency and the cumulative release are closely related to HNTs and Fe_3O_4 contents, crosslinking density and CTS concentration. OFL is released from the magnetic CTS/Fe_3O_4/HNTs/OFL microspheres *via* anomalous transport.

1 Introduction

Chitosan (CTS) is one of the most extensively studied polysaccharides for the preparation of drug delivery carriers.[1,2] CTS microspheres not only have a desirable size in favour of respiration and absorption, but also can provide a high loading efficiency. However, the burst release of drug from pure CTS microspheres is undesirable.[3–5]

Polymer/layered silicate nanocomposites have received significant attentions from both the academic and industrial areas. The introduction of clay minerals into polymeric carriers not only solved the burst release of drugs, but also enhanced the stability in simulated gastric fluid, the mechanical properties and thermal stability of the delivery system.[6–8] For example, Yuan *et al.* reported that the release of doxorubicin hydrochloride from the CTS carrier was prolonged by introducing montmorillonite.[9] Depan's group obtained CTS-*g*-lactic acid/montmorillonite nanohybrids with improved mechanical modulus, strength and release rate of ibuprofen.[10]

Halloysite nanotubes (HNTs), a kind of aluminosilicate mineral, are chemically similar to kaolin, but morphologically have a hollow tubular structure.[11,12] HNTs are promising natural carriers for drug delivery owing to their tubular structure, good biocompatibility and very low cytotoxicity.[13] A very recent study demonstrated that the combination of HNTs and CTS could significantly enhance the tensile properties, storage modulus and glass transition temperature of CTS.[14,15] The Young's modulus and tensile strength of the CTS/HNTs membranes were improved by 21% and 34%, respectively, by incorporating 5% HNTs. Moreover, the CTS/HNTs nanocomposites were cyto-compatible even when the HNTs content was 10%. The introduction of HNTs into CTS films could also improve the bioactivity of the loaded enzyme and greatly promote the direct electron transfer due to their unique tubular structure, high specific surface area and good biocompatibility.[16] The hybrid nanotubes synthesized *via* assembly of CTS onto HNTs could form 3D nanocomposites with hierarchically porous structure, which showed excellent capacity for horseradish peroxidase immobilization through crosslinking by glutaraldehyde.[17]

Magnetic microspheres can be used as carriers for potent drug targeting to a specific site with a powerful magnetic field. Thus, magnetic microspheres show extensive biomedical applications in cell culture and separation, tumor diagnosis, antithrombosis, target pharmaceutical drug design, detection and trapping of viruses, and so on.[18] Magnetic microspheres can be prepared by various methods like *in situ* distillation–precipitation polymerization,[19] layer-by-layer self-assembly,[20] electrospinning method,[21] chemical deposition,[22] suspension polymerization[23] and spray-drying,[24] *etc.* Among them, the spray-drying technique is a one-step constructive process, provides greater control over particle size, particle morphology and powder density, and avoids tedious processes. Hence, the spray-drying technique becomes the most facile and preferred technique for the fabrication of microspheres, and is widely used in medical industry.[25]

Here we present preparation of the magnetic CTS/Fe_3O_4/HNTs/ofloxacin (CTS/Fe_3O_4/HNTs/OFL) microspheres by

Center of Eco-material and Green Chemistry, Lanzhou Institute of Chemical Physics, Chinese Academy of Sciences, Lanzhou, 730000, P. R. China. E-mail: jpzhang@licp.cas.cn; Tel: +86 931 4968251

Published on 02 October 2013. Downloaded by University of Michigan Library on 19/10/2014 06:04:33.

spray-drying for the sustained release of OFL in the gastrointestinal environment. The crosslinking of CTS by glutaraldehyde was *in situ* accomplished in the spray-drying process. The magnetic microspheres were characterized by Fourier transform infrared spectra (FTIR), scanning electron microscopy (SEM), and thermogravimetric analysis (TGA), *etc.* In addition, the encapsulation efficiency (EE), drug loading and *in vitro* release behaviors were systematically evaluated.

2 Materials and methods

2.1 Materials

CTS samples (deacetylation degree, 90%) with number-average molecular weight (M_w) of 3.0×10^5, 6.0×10^5 and 9.0×10^5 were supplied by Zhejiang Yuhuan Ocean Biology Co. (Zhejiang, China). HNTs with an average size of 2.01 μm and a size distribution of 0.044–5.74 μm were purchased from Sigma-Aldrich. Fe_3O_4 nanoparticles (99.5%) with an average size of 0.75 μm and a size distribution of 0.094–1.30 μm were purchased from Aladdin Industrial Corporation (Shanghai, China). OFL was purchased from Jiuzhou pharmaceutical factory (He'nan, China). All the other reagents used were of analytical grade and all solutions were prepared with distilled water.

2.2 Alkali-activation of HNTs

The natural HNTs powder (40.0 g) was suspended in 400 mL of 2 mol L^{-1} NaOH solution. The mixture was sonicated at 50 °C for 1 h, and then washed with distilled water until neutrality. The product was dried at 105 °C to a constant weight and ground into a size of 200 mesh.

2.3 Preparation of magnetic CTS/Fe_3O_4/HNTs/OFL microspheres

Firstly, two suspensions were prepared. A predetermined amount of CTS was dissolved in 350 mL of pH 3.0 acetic acid solution under continuous stirring at 500 rpm, and then an appropriate amount of the Fe_3O_4 nanoparticles were suspended in the solution (suspension I). 0.2 g of OFL was dissolved in 50 mL of pH 3.00 acetic acid solution, and then the alkali-activated HNTs was added under continuous stirring for 12 h to reach the adsorption equilibrium (suspension II). Subsequently, the above-mentioned suspensions (I and II) were mixed under vigorous stirring and the pH was adjusted to 5.0 using 0.1 mol L^{-1} NaOH solution. The mixture was stirred at 500 rpm for 12 h at room temperature. Finally, an appropriate amount of glutaraldehyde was added into the mixture. After stirred for 30 min, the mixture was spray-dried (Mini Spray Dryer, Type SY1600, China) through a 0.7 mm nozzle at a feed rate of 12 mL min^{-1}. The inlet temperature was 150 °C. The pale red magnetic microspheres were dried to a constant weight in an oven at 80 °C.

2.4 Determination of drug loading and EE

The microspheres (0.05 g) were immersed in 50 mL of simulated gastric fluid (pH 1.2) for 24 h until swelling equilibrium, and then sonicated at 50 °C for 1 h to ensure complete release of OFL from the microspheres. Subsequently, the released OFL was separated by centrifugation at 5000 rpm (2653 g) for 20 min. The supernatant after centrifugation was diluted to 200 mL with distilled water. The concentration of OFL was determined using a UV-vis spectrophotometer (SPECORD 200, ANALYTIK JENA AG) at 293 nm. The drug loading and EE were calculated using the following equations.

$$\text{Drug loading (\%)} = c \times v \times 100/m \quad (1)$$

$$\text{EE (\%)} = c \times v \times 100/w \quad (2)$$

where c is the concentration of OFL in the supernatant, g mL^{-1}; v is the volume of the supernatant, mL; m is the mass of magnetic microspheres, g. w is the theoretical weight of OFL in magnetic microspheres, g.

2.5 *In vitro* drug release

In vitro drug release tests were carried out using the USP 30 no. 2 dissolution test apparatus (ZRS-8G, Tianjin University Wireless factory, China) with six rotating paddles. The dialysis bags (MWCO of 8000–14 000 Da) containing the suspension of microspheres (0.1 g) in 5 mL of simulated gastric fluid (pH 1.2) or intestinal fluid (pH 7.4) were placed in 250 mL of the same release medium at 37 °C and stirred at 50 rpm. The simulated gastric fluid and intestinal fluid were prepared using HCl, KH_2PO_4 and NaOH solution according to the Chinese Pharmacopoeia 2005. Aliquots of the solutions were collected at the scheduled intervals and filtered with 0.45 μm-pore nylon filters. After each sample collection, the equal amount of fresh release medium at the same temperature was added back. The filtrate was diluted to 10 mL with the fresh release medium. The concentration of OFL was determined by a UV-vis spectrophotometer at 293 nm. The cumulative release percentage of OFL was determined using eqn (3):

$$\text{OFL cumulative release (\%)} = R_t/L \times 100 \quad (3)$$

where L and R_t represent the initial amount of OFL loaded and cumulative amount of OFL released at time t, respectively.

The release kinetics of OFL were analyzed using the power law.[26,27]

$$M_t/M_\infty = kt^n \quad (4)$$

where M_t/M_∞ is the fractional solute release, t is the release time, k is a constant incorporating structural and geometric characteristics of the device and n is the diffusion exponent characteristic of the release mechanism. According to Siepmann and Peppas's report, for a sphere, $n < 0.43$ represents a Fickian diffusion mechanism, $n > 0.85$ represents a case II transport mechanism, $n = 0.43$–0.85 means both Fickian diffusion and case II transport diffusion, *e.g.*, anomalous transport.

 This journal is © The Royal Society of Chemistry 2013

Published on 02 October 2013. Downloaded by University of Michigan Library on 19/10/2014 06:04:33.

2.6 Characterization

FTIR spectra of the samples were recorded on a Thermo Nicolet NEXUS TM spectrophotometer using KBr pellets. The released OFL in the simulated gastric fluid was also analyzed by FTIR. SEM observations were taken using scanning electron microscopy (JSM-5600LV, JEOL, Ltd) after the samples were dispersed in ethanol and coated by a gold film. The zeta potential was measured using Zetasizer Nano ZS (Malvern Instruments Ltd., Worcestershire, UK) by dispersing 0.1 g of the samples in 10 mL of distilled water. A vibrating-sample magnetometer (VSM) (Lake Shore, 735 VSM, Model 7304, USA) was used at room temperature to characterize the magnetic properties of the Fe_3O_4 nanoparticles and the magnetic CTS/Fe_3O_4/HNTs/OFL microspheres. TGA was carried out using a STA 6000 (PerkinElmer Instrument Co., Ltd. USA) to investigate the thermal stability of the samples (25–800 °C, 10 °C min^{-1}) under nitrogen atmosphere. High-performance liquid chromatogram (HPLC, Waters™ 600 Pump, 2998 Photodiode Array Detector) was used to determine the retention time of OFL using the C_{18} column (4.6 × 150 mm, Waters sunFire™). The mixture of acetonitrile and the buffer solutions (25 : 75, v/v) was used as the mobile phase. OFL was detected at the wavelength of 293 nm. The buffer solution containing 2.0 g ammonium acetate and 3.5 g sodium perchlorate in 650 mL water adjusted with phosphoric acid to a pH 2.2 was prepared according to USP 30.

2.7 Statistical analysis

Statistical analysis for the determination of differences in the adsorption and release characteristics within groups was accomplished using one-way analysis of variance, performed with a statistical program (Origin 8.0). The data were considered to be significantly different at $p < 0.05$. All data are presented as mean values with standard error (mean ± SD). $n = 3$ in each group.

3 Results and discussion

3.1 Preparation of magnetic CTS/Fe_3O_4/HNTs/OFL microspheres

The zeta potentials of CTS, HNTs, Fe_3O_4 and OFL at pH 5.0 solution are shown in Table 1. The zeta potentials of CTS, Fe_3O_4 and OFL in pH 5.0 solution are 50.17, 13.53 and 21.70 mV, respectively, whereas the zeta potential of HNTs is −30.70 mV. So, there is a strong electrostatic interaction of HNTs with CTS, Fe_3O_4 and OFL in their pH 5.0 aqueous suspension. The electrostatic interaction of CTS with HNTs was also confirmed previously according to the FTIR spectra of HNTs and CTS–HNTs.[17] In addition, there is hydrogen bonding between HNTs and CTS owing to their hydrophilic properties besides the electrostatic attraction.[14,17] Many studies have shown that the –CHO groups of glutaraldehyde could couple with the amino groups of CTS to form polymeric networks.[28]

Table 1 The zeta potentials of CTS, HNTs, OFL and Fe_3O_4 in pH 5.0 solution

Samples	CTS	HNTs	OFL	Fe_3O_4
Zeta potentials	50.17 ± 1.54	−30.70 ± 2.14	21.7 ± 2.40	13.52 ± 2.21

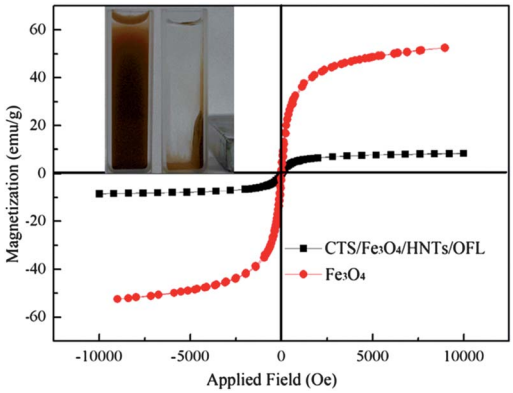

Fig. 1 Magnetic curves of the CTS/Fe_3O_4/HNTs/OFL microspheres and Fe_3O_4 nanoparticles. The microspheres were prepared using 1.0% (w/v) CTS (M_w = 3.0 × 10^5) solution, 10% Fe_3O_4, 10% HNTs and 0.25% glutaraldehyde.

The saturation magnetization of the pure Fe_3O_4 nanoparticles is 52.42 emu g^{-1} at 300 K (Fig. 1). The saturation magnetization of the magnetic microspheres reduces to 8.44 emu g^{-1} because of a large fraction of CTS, HNTs and OFL in the microspheres. The microspheres still can be operated quickly and firmly by a magnet (inset in Fig. 1) indicating nice magnetic responsivity. The superparamagnetic property of the microspheres is critical for their application in biomedical and bioengineering fields, which prevents the magnetic microspheres from aggregation and enables them to redisperse rapidly when a given magnetic field is removed. Moreover, the hysteresis loop of the microspheres shows zero coercivity and zero remanence. This result implies that the microspheres have a superparamagnetic property.

3.2 FTIR analysis

The FTIR spectra of OFL, HNTs, crosslinked CTS and CTS/Fe_3O_4/HNTs/OFL microspheres were shown in Fig. 2. The peaks at 2924 and 2849 cm^{-1} ascribed to the C–H stretching,[17] and 1635 cm^{-1} ascribed to the imine group (C=N) of crosslinked CTS (Fig. 2c)[29] can be seen in the spectrum of the magnetic microspheres. Moreover, the bending vibration of the $-CH_2$ groups at 1375 cm^{-1} and the asymmetric vibration of C–O at 1154 cm^{-1} of crosslinked CTS[15] can also be found in Fig. 2d and e.

The absorption bands of HNTs at 3693 and 3620 cm^{-1} in Fig. 2f are ascribed to the O–H stretching of HNTs.[30] The bands at 536 and 469 cm^{-1} are due to Si–O bending mode and Al–O stretching mode, respectively.[31] After incorporating HNTs into the microsphere, the intensities of the absorption bands at 3693, 536 and 469 cm^{-1} in Fig. 2d and e decrease compared with those of HNTs (Fig. 2f). Moreover, the peak at 3620 cm^{-1} disappears. These changes may be attributed to hydrogen bonding between the –OH of HNTs and the organic groups (–OH and $-NH_2$) of CTS.[17] The broad band of HNTs at 1638 cm^{-1}

Published on 02 October 2013. Downloaded by University of Michigan Library on 19/10/2014 06:04:33.

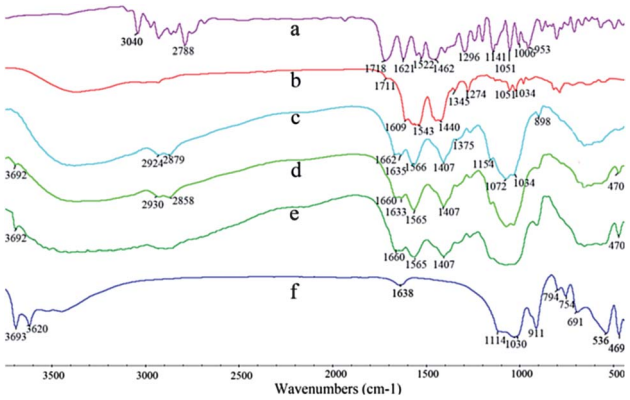

Fig. 2 FTIR spectra of (a) OFL, (b) OFL in the solution (c) crosslinked CTS, (d) CTS/Fe_3O_4/HNTs/OFL (10% HNTs), (e) CTS/Fe_3O_4/HNTs/OFL (20% HNTs) microspheres and (f) HNTs. The microspheres were prepared using 1.0% (w/v) CTS ($M_w = 3.0 \times 10^5$) solution, 10% Fe_3O_4 and 0.25% glutaraldehyde.

is owing to the bending vibration of the adsorbed water. The bands at 1114 and 1030 cm^{-1} are assigned to the apical Si–O and perpendicular Si–O stretching vibration, respectively. The band at 911 cm^{-1} belongs to the bending vibration of the inner –OH. These peaks in Fig. 2d and e are broadened, which could overlap the absorption peaks at 1635, 1071, 1034 and 898 cm^{-1} of crosslinked CTS.

In the FTIR spectrum of OFL (Fig. 2a), the prominent characteristic peak at 3040 cm^{-1} belongs to stretching vibration of –OH group and intramolecular hydrogen bonding. The peaks at 1718 and 1621 cm^{-1} are assigned to the C=O stretching and N–H bending vibration, respectively. The bands at 1522 and 1462 cm^{-1} are due to the $-CH_2$ stretching vibration of the aromatic ring and the $-CH_2$ stretching vibration of benzoxazine ring, respectively. The strong band at 1296 cm^{-1} is ascribed to the stretching vibration of the C–O–C group. In addition, the strong absorption peak at 1051 cm^{-1} is attributed to the C–F group. The position of these peaks is in accordance with the previous report.[32] However, the characteristic peaks of OFL are covered by the absorption bands of HNTs and crosslinked CTS, and cannot be seen in Fig. 2d and e.

To prove the successful entrapment of OFL, the microspheres were immersed in pH 1.2 solution for 12 h and the solution was analyzed by FTIR. As can be seen from Fig. 2b, the characteristic bands of OFL at 1711, 1609, 1440 and 1274 cm^{-1} are detected. This result demonstrates that OFL is successfully entrapment in the magnetic microspheres and can be safely released in the simulated gastric fluid. The above-mentioned solution was further analyzed by HPLC to prove that the spray-drying process has no influence on the molecular structure of OFL. As can be seen from Fig. 3, the retention time of pure OFL is 3.87 min and that of OFL released from the microspheres is 3.82 min. This result suggests that OFL is stable during the spray-drying process without any chemical deterioration of its functional groups.

3.3 Morphological analysis

The surface morphology of CTS/OFL, CTS/Fe_3O_4/OFL, CTS/HNTs/OFL and CTS/Fe_3O_4/HNTs/OFL microspheres were observed by SEM (Fig. 4). HNTs and Fe_3O_4 nanoparticles have influences on surface morphology of the microspheres. The surface of the CTS/OFL microspheres (Fig. 4a) is very smooth. The shrunk and collapsed microspheres are formed (Fig. 4b) when Fe_3O_4 nanoparticles are introduced into the CTS/OFL microspheres. This result is similar with the previous work that the magnetic CTS/iron (II, III) oxide nanoparticles prepared by spray-drying had a quite coarse surface and irregular spheroid,[24] but different from the result of Denkbaş *et al.*[33] who prepared CTS/Fe_3O_4 magnetic microspheres with a rather smooth surface by the suspension crosslinking technique. The surface becomes rougher compared with the CTS/OFL microspheres when HNTs are incorporated into CTS/OFL microspheres (Fig. 4c). The surface morphology of the CTS/Fe_3O_4/HNTs/OFL microspheres (Fig. 4d) is very similar to that of the CTS/Fe_3O_4/OFL microspheres when HNTs and Fe_3O_4 nanoparticles are simultaneously introduced. The above-mentioned results indicate that Fe_3O_4 nanoparticles have remarkably effect on shape of the

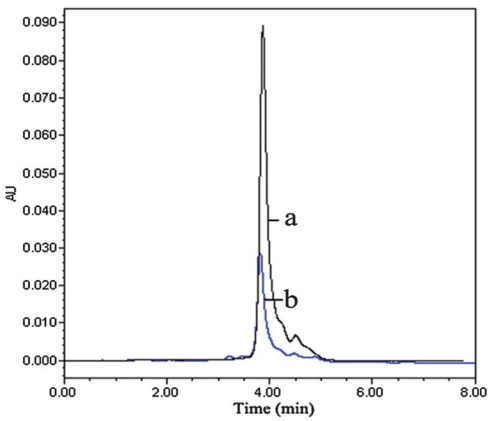

Fig. 3 HPLC chromatograms of (a) pure OFL and (b) OFL released from the magnetic microspheres. The magnetic microspheres were prepared using 1.0% (w/v) CTS ($M_w = 3.0 \times 10^5$) solution, 10% Fe_3O_4, 10% HNTs and 0.25% glutaraldehyde.

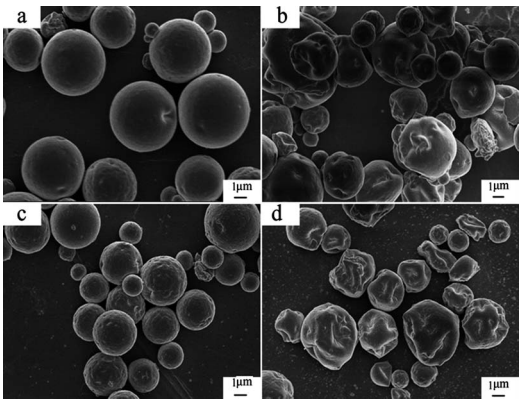

Fig. 4 SEM of (a) CTS/OFL, (b) CTS/Fe_3O_4/OFL (10% Fe_3O_4), (c) CTS/HNTs/OFL (10% HNTs) and (d) CTS/Fe_3O_4/HNTs/OFL microspheres (10% Fe_3O_4 and 10% HNTs). The microspheres were prepared using 1.0% (w/v) CTS ($M_w = 3.0 \times 10^5$) solution and 0.25% glutaraldehyde.

 This journal is © The Royal Society of Chemistry 2013

Published on 02 October 2013. Downloaded by University of Michigan Library on 19/10/2014 06:04:33.

CTS/Fe_3O_4/HNTs/OFL microspheres, whereas HNTs only affect the roughness.

3.4 Thermal stability

The decomposition of the microspheres takes place in two main steps according to the DSC curves (insert in Fig. 5). The first endothermic peak at around 75 °C is attributed to the dehydration of physisorbed water and interlayer water.[11,34] The second exothermic peak at about 275 °C is attributed to breaking of C–O–C glycosidic bonds of CTS.[29]

TGA curves of the microspheres were shown in Fig. 5. As can be seen, the weight loss rate of the CTS/Fe_3O_4/OFL microspheres is lower than that of the CTS/Fe_3O_4/HNTs/OFL microspheres at temperature below 150 °C. This is because of the physisorbed water and interlayer water of the introduced HNTs. The weight loss rate of the microspheres increases with increasing the HNTs content to 10%, and then decreases with further increasing the HNTs content from 20 to 40% when the temperature is above 250 °C. This result indicates that the incorporation of a higher content of HNTs is helpful to improve thermal stability of the microspheres because HNTs act as heat barriers. The heat barriers make the small molecules generated during the thermal decomposition difficult to permeate and volatilize.[35]

3.5 Adsorption and release for OFL

3.5.1 Effect of M_w of CTS. The effects of M_w of CTS on EE, drug loading and cumulative release of OFL from the microspheres were shown in Table 2 and Fig. 6. The drug loading and EE of the microspheres decrease with increasing M_w of CTS from 3.0×10^5 to 6.0×10^5, and then increase with further increasing M_w to 9.0×10^5. However, the cumulative releases show an opposite tendency with increasing M_w of CTS (Fig. 6).

It is conceived that an increase in M_w of CTS would increase the viscosity of the CTS acidic aqueous solution. Thus, the increase in M_w of CTS is expected to increase the EE.[36] The increase in EE and drug loading with M_w of CTS from 6.0×10^5 to 9.0×10^5 can be well explained by the increased viscosity of the CTS solution. However, the changing tendency of EE is not always accordant with the M_w of CTS. This may be because EE is not only related to the viscosity of CTS solution, but also to the crosslinking density of the polymeric network. CTS solution of smaller M_w has lower viscosity, which would facilitate the

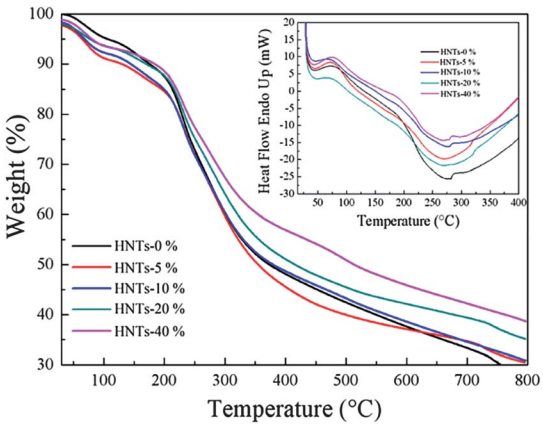

Fig. 5 TGA and DSC curves of the microspheres with different HNTs content. The microspheres were prepared using 1.0% (w/v) CTS ($M_w = 3.0 \times 10^5$) solution, 10% Fe_3O_4 and 0.25% glutaraldehyde.

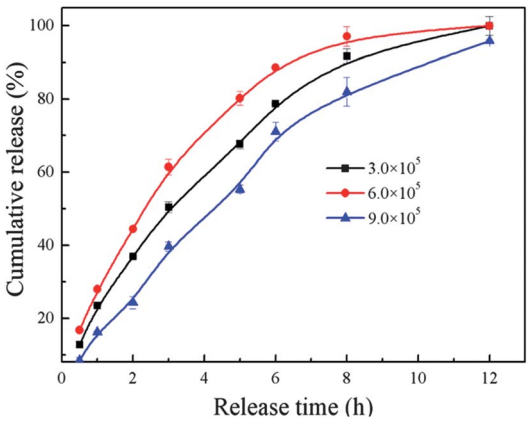

Fig. 6 Variation of cumulative release of OFL from the microspheres with M_w of CTS in pH 1.2 solution. The microspheres were prepared using 1.0% (w/v) CTS solution, 10% Fe_3O_4, 10% HNTs and 0.25% glutaraldehyde.

Table 2 Variation of drug loading, encapsulation and release of OFL with M_w of CTS, CTS concentration and HNTs content

Parameters		Drug loading (%)	EE (%)	Yield (%)	Power law n	Power law k	Power law r^2	Release mechanism
M_w of CTS ($\times 10^5$)	3.0	2.94 ± 0.10	72.51 ± 0.65	42.2	0.702	22.23	0.995	Anomalous
	6.0	2.77 ± 0.18	68.76 ± 1.21	38.9	0.646	27.67	0.992	Anomalous
	9.0	3.02 ± 0.16	74.98 ± 1.09	39.8	0.822	15.24	0.994	Anomalous
CTS conc. (%)	0.5	4.24 ± 0.05	63.08 ± 1.07	45.2	0.835	24.27	0.989	Anomalous
	1.0	2.94 ± 0.04	72.51 ± 2.39	42.2	0.702	22.23	0.995	Anomalous
	1.5	2.48 ± 0.10	85.57 ± 1.34	44.1	0.579	19.54	0.997	Anomalous
Content of HNTs (%)	0	3.23 ± 0.03	73.67 ± 3.01	45.2	0.944	17.78	0.988	Case II
	5	3.00 ± 0.19	70.90 ± 3.14	43.2	0.700	18.24	0.978	Anomalous
	10	2.94 ± 0.21	72.51 ± 2.72	42.2	0.702	22.23	0.995	Anomalous
	20	2.75 ± 0.07	73.71 ± 2.70	43.6	0.694	33.19	0.990	Anomalous
	40	2.43 ± 0.04	74.95 ± 2.41	43.5	0.635	38.19	0.983	Anomalous

Published on 02 October 2013. Downloaded by University of Michigan Library on 19/10/2014 06:04:33.

reaction between CTS and glutaraldehyde, and then enhance the crosslinking density.[37] Consequently, higher EE and drug loading were detected when CTS with a M_w of 3.0×10^5 was used in comparison with that with a M_w of 6.0×10^5. Thus, the detected EE and drug loading with increasing the M_w of CTS are the results of the synergistic effects of viscosity of CTS and the crosslinking density.

The synergistic effects of viscosity of CTS and the crosslinking density by changing M_w of CTS also have evident influences on the cumulative release rate of OFL. The more compact network is not conducive to the emigration and release of OFL from the microspheres. Therefore, CTS with a M_w of 3.0×10^5 decreases the cumulative release rate of OFL compared with CTS with a M_w of 6.0×10^5. On the other hand, the higher viscosity of the CTS solution with M_w of 9.0×10^5 limits the free movement of loaded OFL molecules, and then decreases the cumulative release rate of OFL. It also can be seen from Table 2 that n is in the range of 0.646 to 0.822, indicating that the release of OFL is due to the anomalous transport.

3.5.2 Effect of CTS concentration. The EE and drug loading of the magnetic microspheres with different CTS concentration were shown in Table 2. The drug loading decreases and the EE increases with increasing the CTS concentration. In addition, the cumulative release of OFL in pH 1.2 solution decreases with increasing the CTS concentration (Fig. 7). It is known that the amino groups of CTS could react with CHO groups of glutaraldehyde *via* the Schiff base reaction to form polymeric network. The crosslinking density of the polymeric network increases with increasing CTS concentration, which restricts emigration of OFL from the microspheres, and then increases EE and decreases the release of OFL. In addition, an increase in the viscosity of CTS with increasing CTS concentration also prevents the free movement of OFL molecules in the microspheres, which are also responsible for the increase of the EE. It can be seen from Table 2 that n is in the range of 0.579 to 0.835, indicating that the release of OFL is owing to the anomalous transport.

3.5.3 Effect of HNTs content. The effects of HNTs content on EE and drug loading of OFL were shown in Table 2. The drug loading decreases with increasing the HNTs content. The EE decreases firstly, and then increases with increasing the HNTs content, but there are no significant differences in EE with increasing the HNTs content ($p > 0.05$). The tendency may be attributed to the following facts. CTS can react with glutaraldehyde to form a compact network when the HNTs content is 0%, which is of benefit to the encapsulation of OFL in the microspheres. When HNTs are introduced into the CTS solution, the tubular HNTs may hinder the Schiff base reaction between CTS and glutaraldehyde. Consequently, a loose network was formed, which results in the decrease in EE. The increase in EE with increasing the HNTs content from 5 to 40% is because more OFL is adsorbed onto HNTs *via* electrostatic interaction. The synergistic effects of the electrostatic interaction and the crosslinking density make no significant change in EE with increasing the HNTs content.

Fig. 8 shows the effect of HNTs content on the cumulative release of OFL. The cumulative release of OFL decreases firstly, and then increases with increasing the HNTs content. This tendency is consistent with that of EE. This is attributed to the role that HNTs play in the microspheres. As has been reported by Lvov *et al.*, HNTs could be used as vehicles for the slow-release of nifedipine, furosemide and dexamethasone.[38] Thus, OFL adsorbed on HNTs needs to migrate from HNTs to the CTS/Fe_3O_4/HNTs/OFL network firstly, and then from the network to the buffer solution. There is a longer path for OFL to migrate from the CTS/Fe_3O_4/HNTs/OFL microspheres to the buffer solution compared with the CTS/Fe_3O_4/OFL microspheres. Consequently, the decrease in cumulative release of OFL was observed when 5% HNTs was introduced. The crosslinking density decreases with further increasing HNTs content to 40% because HNTs could hinder the reaction between CTS and glutaraldehyde. Thus, an increase in the cumulative release of OFL was observed. In addition, the rougher surface of the CTS/Fe_3O_4/HNTs/OFL microspheres is also of benefit to the release of OFL compared to CTS/Fe_3O_4/OFL. The value of n for the CTS/Fe_3O_4/OFL microspheres is 0.944 (above 0.85, Table 2), indicating that the release of OFL is owing to case II transport, namely swelling-controlled. When HNTs are incorporated, the

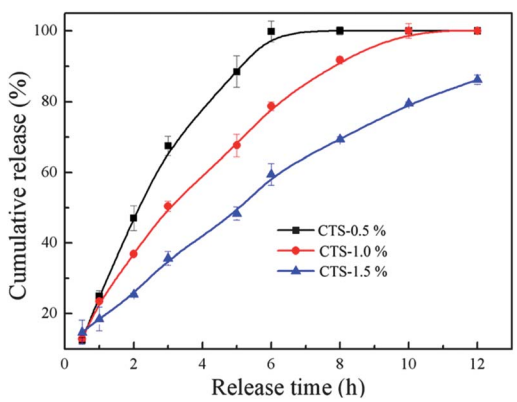

Fig. 7 Variation of cumulative release of OFL from the microspheres with CTS concentration in pH 1.2 solution. The microspheres were prepared using CTS with a M_w of 3.0×10^5, 10% Fe_3O_4, 10% HNTs and 0.25% glutaraldehyde.

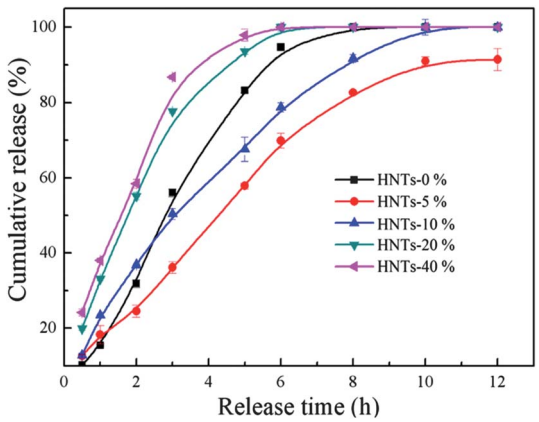

Fig. 8 Variation of cumulative release of OFL from the microspheres with HNTs content in pH 1.2 solution. The microspheres were prepared using 1.0% (w/v) CTS ($M_w = 3.0 \times 10^5$) solution, 10% Fe_3O_4 and 0.25% glutaraldehyde.

 This journal is © The Royal Society of Chemistry 2013

Published on 02 October 2013. Downloaded by University of Michigan Library on 19/10/2014 06:04:33.

value of n is in the range of 0.635 to 0.702, which means the release is controlled by both diffusion swelling, namely anomalous transport.

3.5.4 Effect of Fe_3O_4 content. The effects of Fe_3O_4 content on the EE, drug loading and cumulative release were investigated (Table 3 and Fig. 9). The drug loading decreases with increasing the Fe_3O_4 content as shown in Table 3. No evident change in EE can be seen when the Fe_3O_4 content is less than 10%, and then decreases to 72.39 ± 1.79% with further increasing the Fe_3O_4 content to 20%. The cumulative release has the same tendency with the drug loading and decreases with increasing the Fe_3O_4 content (Fig. 9). There is no electrostatic interaction between OFL and Fe_3O_4 according to their zeta potentials (Table 1). The Fe_3O_4 nanoparticles act as fillers and prevent OFL emigrating from the microspheres, which is responsible for the increase in EE and the decrease in the cumulative release. In addition, the Fe_3O_4 nanoparticles makes the microspheres shrink and collapse, which could also contribute to the decrease of the cumulative release rate of OFL compared to the CTS/HNTs/OFL microspheres. However, the excess Fe_3O_4 nanoparticles count against the encapsulation of OFL because of the electrostatic repulsion between them. As a result, the EE decreases when the Fe_3O_4 content increases to 20%. As can be seen from Table 3, n is in the range of 0.516 to 0.664 for the microspheres with different Fe_3O_4 content, indicating that the release of OFL is owing to anomalous transport.

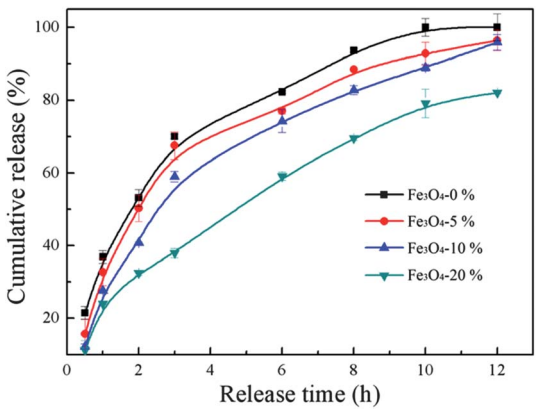

Fig. 9 Variation of cumulative release of OFL from the magnetic microspheres with Fe_3O_4 content in pH 1.2 solution. The microspheres were prepared using 1.0% (w/v) CTS ($M_w = 3.0 \times 10^5$) solution, 10% HNTs and 0.25% glutaraldehyde.

3.5.5 Effect of glutaraldehyde concentration. The EE and drug loading of the microspheres with different glutaraldehyde concentration are shown in Table 3. The drug loading firstly decreases, and then increases with increasing the glutaraldehyde concentration. The variation of EE is accordant with that of the drug loading. No significant difference in EE is detected with increasing the glutaraldehyde concentration from 0.125 to 0.375% ($p > 0.05$).

The cumulative release of OFL decreases with increasing glutaraldehyde concentration, especially when the glutaraldehyde concentration reaches 0.250% (Fig. 10). Glutaraldehyde acts as the crosslinker in forming the microspheres by reacting with the $-NH_2$ groups of CTS. The microspheres are only formed *via* electrostatic interaction and hydrogen bonding without glutaraldehyde. Thus, very fast release of OFL was observed. The microspheres are chemically crosslinked when glutaraldehyde is introduced and the crosslinking density increases with increasing the glutaraldehyde concentration. Consequently, the diffusion of the loaded OFL into the dissolution media is slowed down.[28,29] As can be seen from Table 3, n is in the range of 0.461 to 0.705 for the microspheres with different glutaraldehyde concentration, indicating that the release of OFL is owing to anomalous transport.

3.5.6 Effect of pH. The effect of pH on the release of OFL from the microspheres is shown in Fig. 11. The cumulative

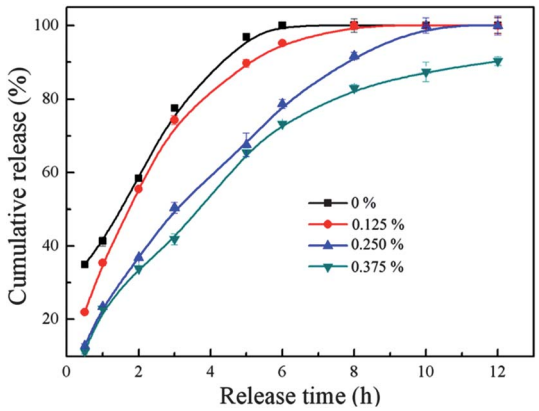

Fig. 10 Variation of cumulative release of OFL from the microspheres with glutaraldehyde concentration in pH 1.2 solution. The microspheres were prepared using 1.0% (w/v) CTS ($M_w = 3.0 \times 10^5$) solution, 10% Fe_3O_4 and 10% HNTs.

Table 3 Variation of drug loading, encapsulation and release of OFL with Fe_3O_4 content and glutaraldehyde concentration

Parameters		Drug loading (%)	EE (%)	Yield (%)	Power law n	k	r^2	Release mechanism
Content of Fe_3O_4 (%)	0	3.34 ± 0.17	75.32 ± 1.32	45.1	0.516	34.67	0.966	Anomalous
	5	3.04 ± 0.15	77.87 ± 1.91	44.2	0.595	29.17	0.957	Anomalous
	10	2.23 ± 0.01	76.75 ± 0.65	42.2	0.664	23.88	0.952	Anomalous
	20	2.12 ± 0.05	72.39 ± 1.79	45.1	0.614	20.00	0.971	Anomalous
Glutaraldehyde conc. (%)	0	4.16 ± 0.11	97.89 ± 2.15	45.6	0.461	44.77	0.975	Anomalous
	0.125	3.04 ± 0.21	77.04 ± 2.95	44.8	0.599	34.99	0.988	Anomalous
	0.250	3.14 ± 0.32	77.87 ± 1.91	42.2	0.702	22.23	0.995	Anomalous
	0.375	3.23 ± 0.06	79.53 ± 0.31	46.7	0.705	20.28	0.989	Anomalous

Published on 02 October 2013. Downloaded by University of Michigan Library on 19/10/2014 06:04:33.

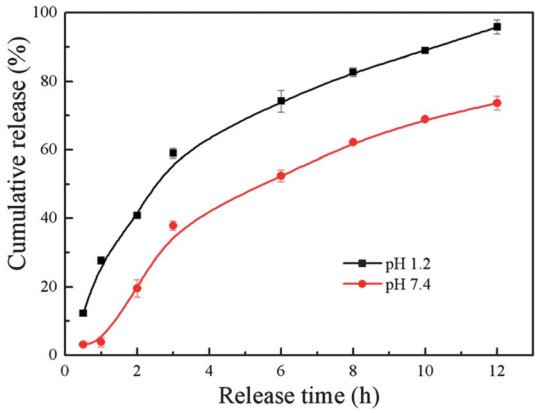

Fig. 11 Cumulative release of OFL from the microspheres in pH 1.2 and 7.4 solutions. The microspheres were prepared using 1.0% (w/v) CTS ($M_w = 3.0 \times 10^5$) solution, 10% Fe_3O_4, 10% HNTs and 0.25% glutaraldehyde.

release of OFL at pH 1.2 is above 90%, whereas that at pH 7.4 is below 80% in 12 h. This difference is attributed to the following facts. (i) The solubility of OFL at different pH. OFL molecules (pK_{a1} 6.08 and pK_{a2} 8.25) exist as cationic, neutral and anionic species in acidic, neutral and alkaline solutions, respectively.[39] OFL molecules exist as cationic species when pH < 3.16, cationic and neutral species when pH 3.16–6.96, neutral species when pH 6.96–7.22, neutral and anionic species when pH 7.22–9.12, and anionic species when pH > 9.12.[40] Therefore, the solubility of OFL is high in pH 1.2 solution, whereas that in pH 7.4 solution is low. As a result, it is easy for OFL to release from the magnetic microspheres at pH 1.2, which leads to a higher cumulative release compared with that at pH 7.4. (ii) The solubility of CTS at different pH. CTS is soluble in acidic aqueous solution, but insoluble in neutral and basic solutions. Therefore, the dissolution of uncrosslinked and partly crosslinked CTS from the microspheres in pH 1.2 solution can accelerate the release of OFL. (iii) The interaction between OFL and the microspheres. The isoelectric point of HNTs is around pH 3.[41] Both HNTs and OFL are positively charged at pH 1.2. Therefore, the electrostatic repulsion between HNTs and OFL in pH 1.2 should also contribute to the higher cumulative release of OFL.

4 Conclusions

The magnetic CTS/Fe_3O_4/HNTs/OFL microspheres with sustained release properties in the simulated gastrointestinal fluid were prepared by a facile spray-drying technique. The magnetic microspheres possess superparamagnetic property and fast magnetic response. The Fe_3O_4 nanoparticles have remarkable effect on morphology of the microspheres, whereas HNTs only affect roughness of the surface. The introduction of HNTs changes the release behavior of OFL from the case II transport to the anomalous transport. The EE and cumulative release are related to both the crosslinking density and the viscosity of CTS solution. The electrostatic interaction and hydrogen bonding between HNTs and CTS could influence the crosslinking density of CTS with glutaraldehyde, which results in an increase in the cumulative release with increasing HNTs content from 10 to 40%. An appropriate amount of the Fe_3O_4 nanoparticles has no effect on EE, but could prolong the release of OFL. The magnetic microspheres are promising to be used as carriers for magnetically driven targeted delivery of drugs in the gastrointestinal tract.

Acknowledgements

We are grateful for financial support of the National Natural Science Foundation of China (no. 51003112) and the "Hundred Talents Program" of the Chinese Academy of Sciences.

Notes and references

1 T. Kean and M. Thanou, *Adv. Drug Delivery Rev.*, 2010, **62**, 3–11.
2 N. Bhattarai, J. Gunn and M. Q. Zhang, *Adv. Drug Delivery Rev.*, 2010, **62**, 83–99.
3 L. Ochiuz and J. E. Peris, *Med. Chem.*, 2009, **5**, 191–196.
4 P. Giunchedi, C. Juliano, E. Gavini, M. Cossu and M. Sorrenti, *Eur. J. Pharm. Biopharm.*, 2002, **53**, 233–239.
5 P. He, S. S. Davis and L. Illum, *Int. J. Pharm.*, 1999, **187**, 53–65.
6 J. P. Zhang, Q. Wang and A. Q. Wang, *Acta Biomater.*, 2010, **6**, 445–454.
7 N. Banik, A. Hussain, A. Ramteke, H. K. Sharmac and T. K. Maji, *RSC Adv.*, 2012, **2**, 10519–10528.
8 Q. Wang, X. L. Xie, X. W. Zhang, J. P. Zhang and A. Q. Wang, *Int. J. Biol. Macromol.*, 2010, **46**, 356–362.
9 Q. Yuan, J. Shah, S. Hein and R. D. K. Misra, *Acta Biomater.*, 2010, **6**, 1140–1148.
10 D. Depan, A. P. Kumar and R. P. Singh, *Acta Biomater.*, 2009, **5**, 93–100.
11 E. Joussein, S. Petit, J. Churchman, B. Theng, D. Righi and B. Delvaux, *Clay Miner.*, 2005, **40**, 383–426.
12 W. O. Yah, A. Takahara and Y. M. Lvov, *J. Am. Chem. Soc.*, 2012, **134**, 1853–1859.
13 V. Vergaro, E. Abdullayev, Y. M. Lvov, A. Zeitoun, R. Cingolani, R. Rinaldi and S. Leporatti, *Biomacromolecules*, 2010, **11**, 820–826.
14 M. Liu, Y. Zhang, C. Wu, S. Xiong and C. Zhou, *Int. J. Biol. Macromol.*, 2012, **51**, 566–575.
15 R. T. De Silva, P. Pasbakhsh, K. L. Goh, S. P. Chai and H. Ismail, *Polym. Test.*, 2013, **32**, 265–271.
16 X. Sun, Y. Zhang, H. Shen and N. Jia, *Electrochim. Acta*, 2010, **56**, 700–705.
17 R. Zhai, B. Zhang, Y. Z. Wan, C. C. Li, J. T. Wang and J. D. Liu, *Chem. Eng. J.*, 2013, **214**, 304–309.
18 M. Arruebo, R. Fernández-Pacheco, M. R. Ibarra and J. Santamaría, *Nano Today*, 2007, **2**, 22–32.
19 B. Liu, W. Zhang, F. K. Yang, H. L. Feng and X. L. Yang, *J. Phys. Chem. C*, 2011, **115**, 15875–15884.
20 P. C. Du, J. Zeng, B. Mu and P. Liu, *Mol. Pharmaceutics*, 2013, **10**, 1705–1715.
21 K. Lai, W. Jiang, J. Z. Tang, Y. Wu, B. He, G. Wang and Z. W. Gu, *RSC Adv.*, 2012, **2**, 13007–13017.
22 X. G. Yu, J. Q. Wan, Y. Shan, K. Z. Chen and X. D. Han, *Chem. Mater.*, 2009, **21**, 4892–4898.

 This journal is © The Royal Society of Chemistry 2013

Published on 02 October 2013. Downloaded by University of Michigan Library on 19/10/2014 06:04:33.

23 Y. H. Chen, M. Franzre, R. H. Lin, L. L. Chen, C. Y. Chang, Y. H. Yu and P. C. Chiang, *Ind. Eng. Chem. Res.*, 2009, **48**, 7616–7623.
24 H. Y. Huang, Y. T. Shieh, C. M. Shih and Y. K. Twu, *Carbohydr. Polym.*, 2010, **81**, 906–910.
25 P. Tewa-Tagne, S. Briançon and H. Fessi, *Eur. J. Pharm. Sci.*, 2007, **30**, 124–135.
26 P. L. Ritger and N. A. Peppas, *J. Controlled Release*, 1987, **5**, 23–36.
27 J. Siepmanna and N. A. Peppas, *Adv. Drug Delivery Rev.*, 2001, **48**, 139–157.
28 F. L. Mi, C. Y. Kuan, S. S. Shyu, S. T. Lee and S. F. Chang, *Carbohydr. Polym.*, 2000, **41**, 389–396.
29 V. H. Kulkarni, P. V. Kulkarni and J. Keshavayya, *J. Appl. Polym. Sci.*, 2007, **103**, 211–217.
30 R. L. Frost, J. Kristof, E. Horvath and J. T. Kloprogge, *J. Colloid Interface Sci.*, 2000, **226**, 318–327.
31 Y. Deng, G. N. White and J. B. Dixon, *J. Colloid Interface Sci.*, 2002, **250**, 379–393.
32 S. Sahoo, C. K. Chakraborti and P. K. Behera, *Int. J. PharmTech Res.*, 2012, **4**, 382–391.
33 E. B. Denkbaş, E. Kiliçay, C. Birlikseven and E. Öztürk, *React. Funct. Polym.*, 2002, **50**, 225–232.
34 S. Kadi, S. Lellou, K. Marouf-Khelifa, J. Schott, I. Gener-Batonneau and A. Khelifa, *Microporous Mesoporous Mater.*, 2012, **158**, 47–54.
35 Y. A. Zheng and A. Q. Wang, *J. Macromol. Sci., Part A: Pure Appl. Chem.*, 2010, **47**, 33–38.
36 X. G. Chen, L. Zheng, Z. Wang, C. Y. Lee and H. J. Park, *J. Agric. Food Chem.*, 2002, **50**, 5915–5918.
37 J. P. Zhang, Q. Wang and A. Q. Wang, *Carbohydr. Polym.*, 2007, **68**, 367–374.
38 Y. M. Lvov, D. G. Shchukin, H. Möhwald and R. R. Price, *ACS Nano*, 2008, **2**, 814–820.
39 K. W. Goyne, J. Chorover, J. D. Kubicki, A. R. Zimmerman and S. L. Brantley, *J. Colloid Interface Sci.*, 2005, **283**, 160–170.
40 Q. L. Huang, Y. P. Deng and S. Y. Xu, *China Pharm.*, 2010, **21**, 3907–3909.
41 L. Guimaraês, A. N. Enyashin, G. Seifert and H. A. Duarte, *J. Phys. Chem. C*, 2010, **114**, 11358–11363.